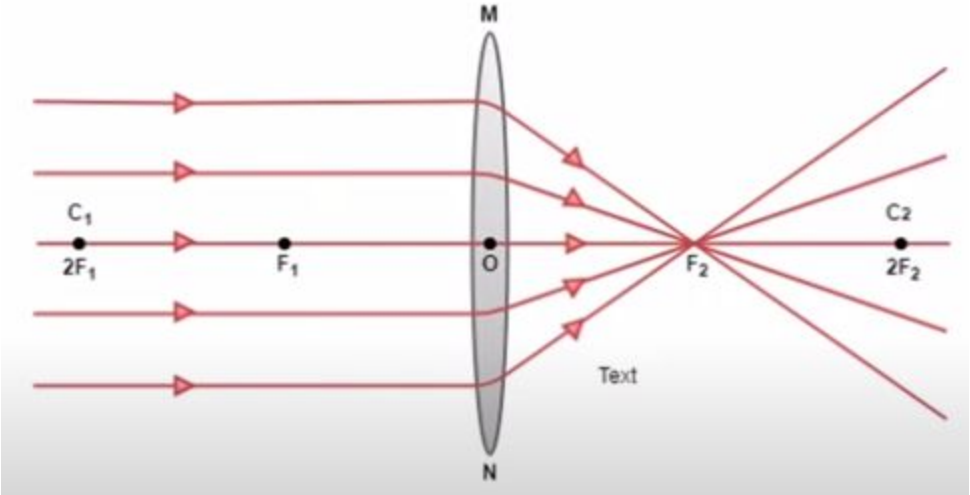
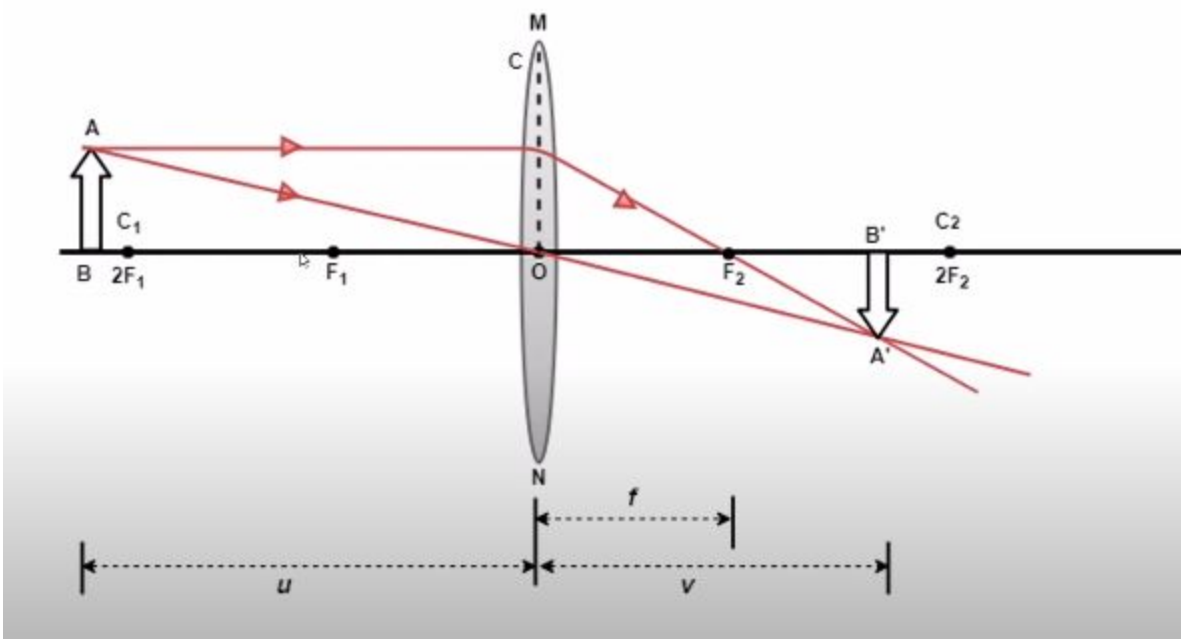
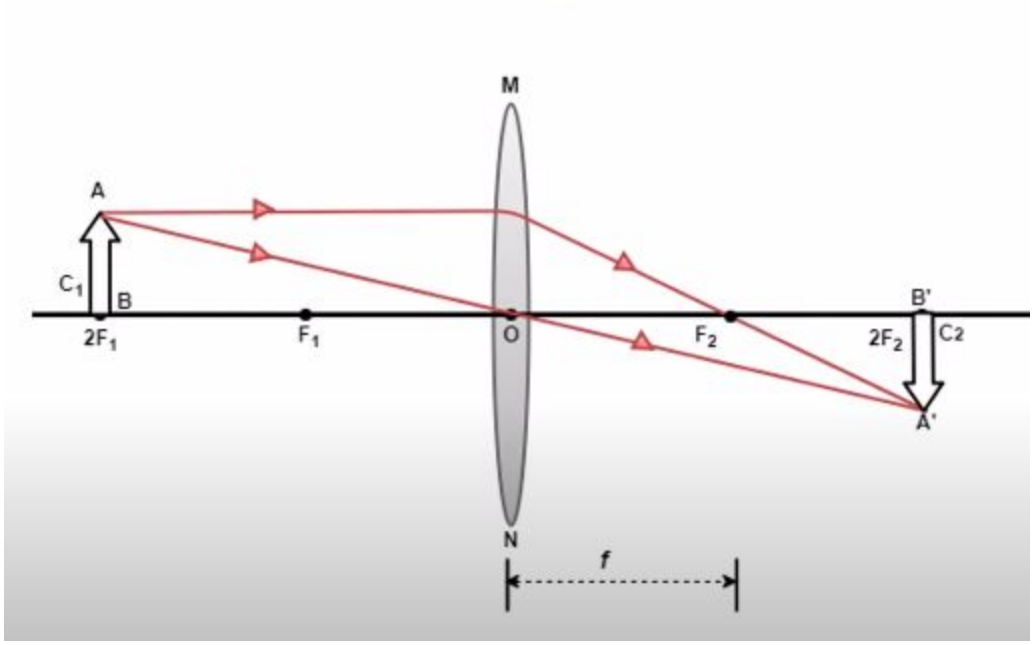
Object at infinity



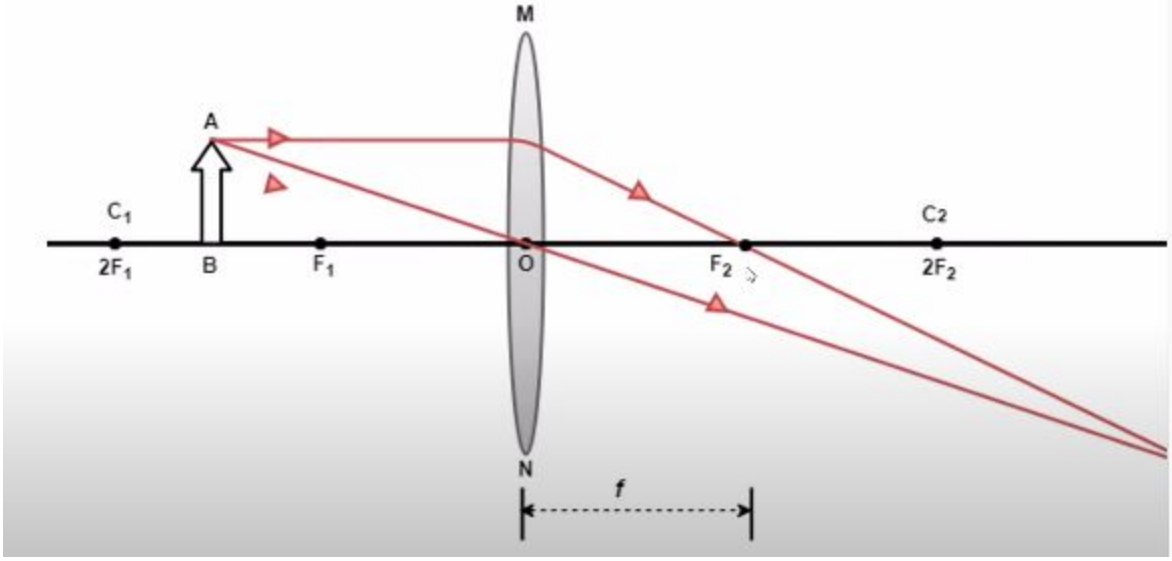
$\infty > u > 2f$



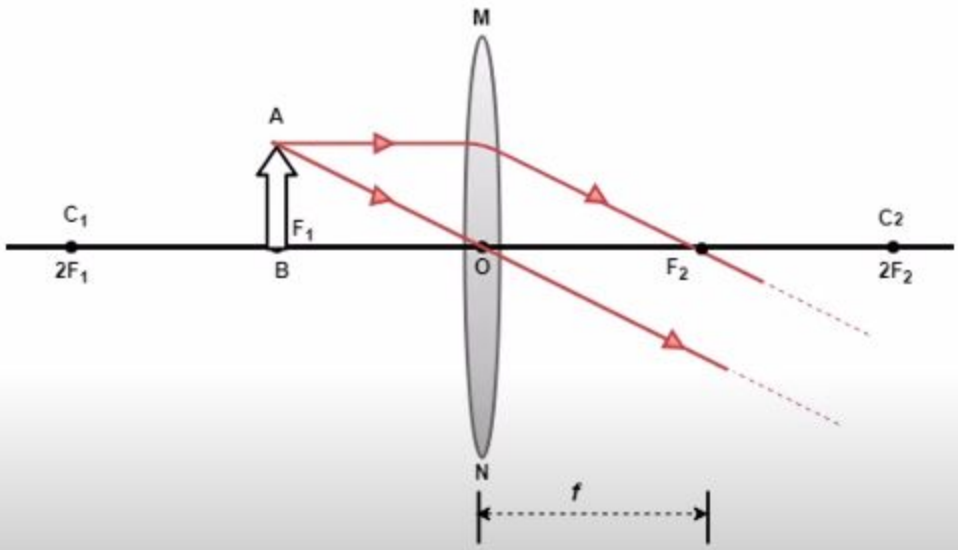
u = 2f



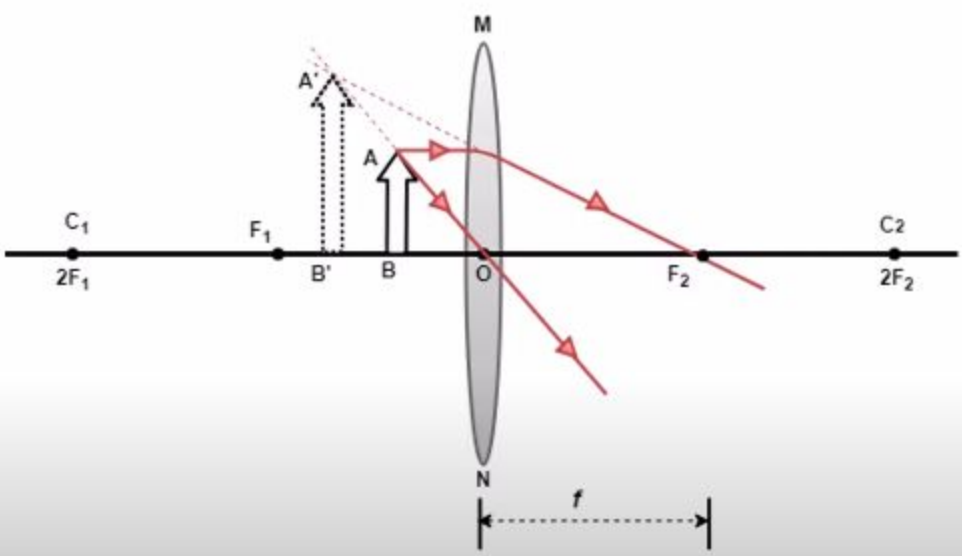
2f > u > f



u = f



u < f



Homework

- Arrive at the imaging conditions for
  - Concave lens
  - Mirrors (concave and convex)
- Derive the Newtonian form of the the lens equation $x_i x_0 = f^2$

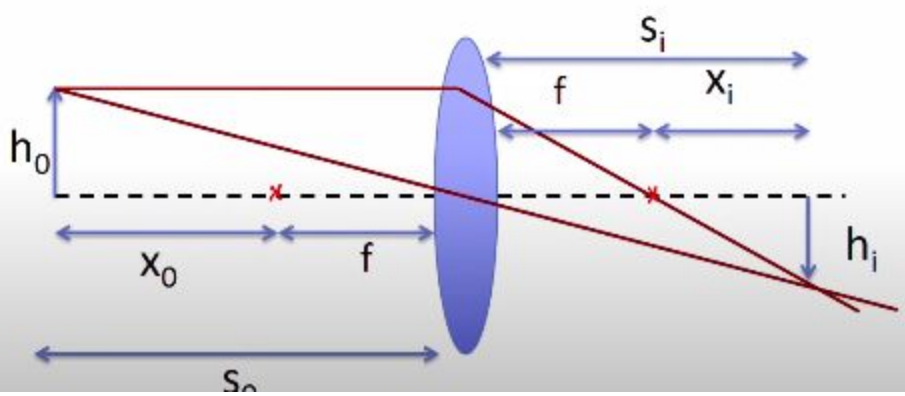
$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

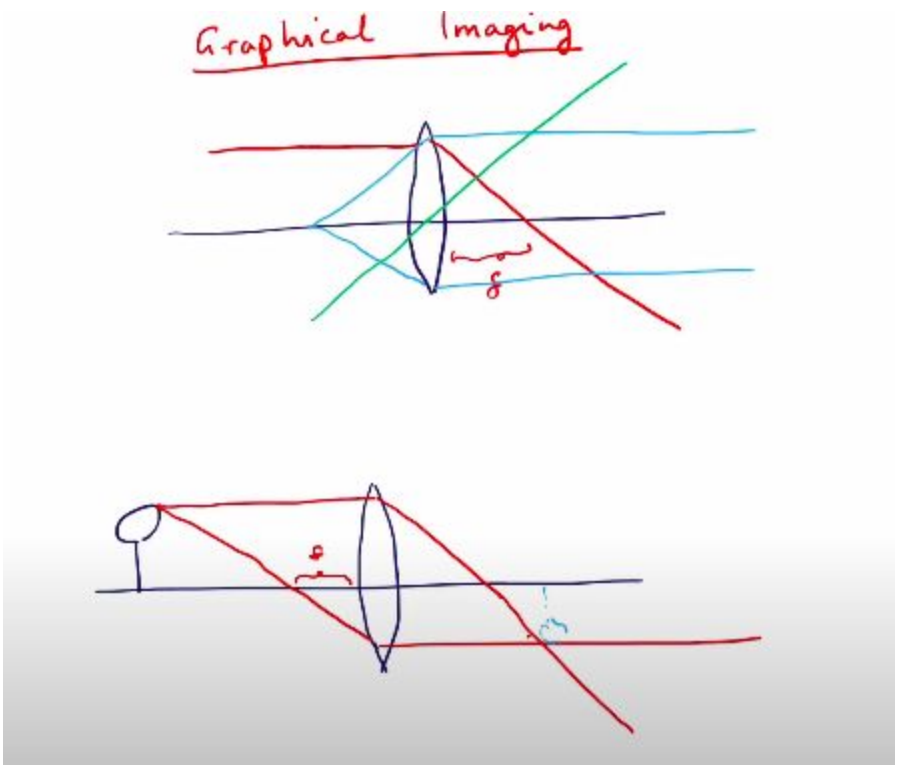
Transverse Magnification

$$M = \frac{h_i}{h_0} = -\frac{s_i}{s_0} = -\frac{x_i}{f} = \frac{f}{-x_0}$$

$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$

$\frac{1}{s_0} + \frac{1}{s_i} = \frac{1}{f}$





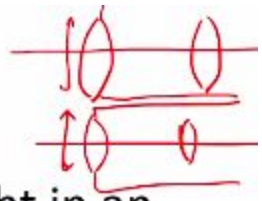
Apertures

Apertures and Stops

- A stop is anything that limits which light rays can get through an optical system
  - Eg: single lens only passes rays within its diameter

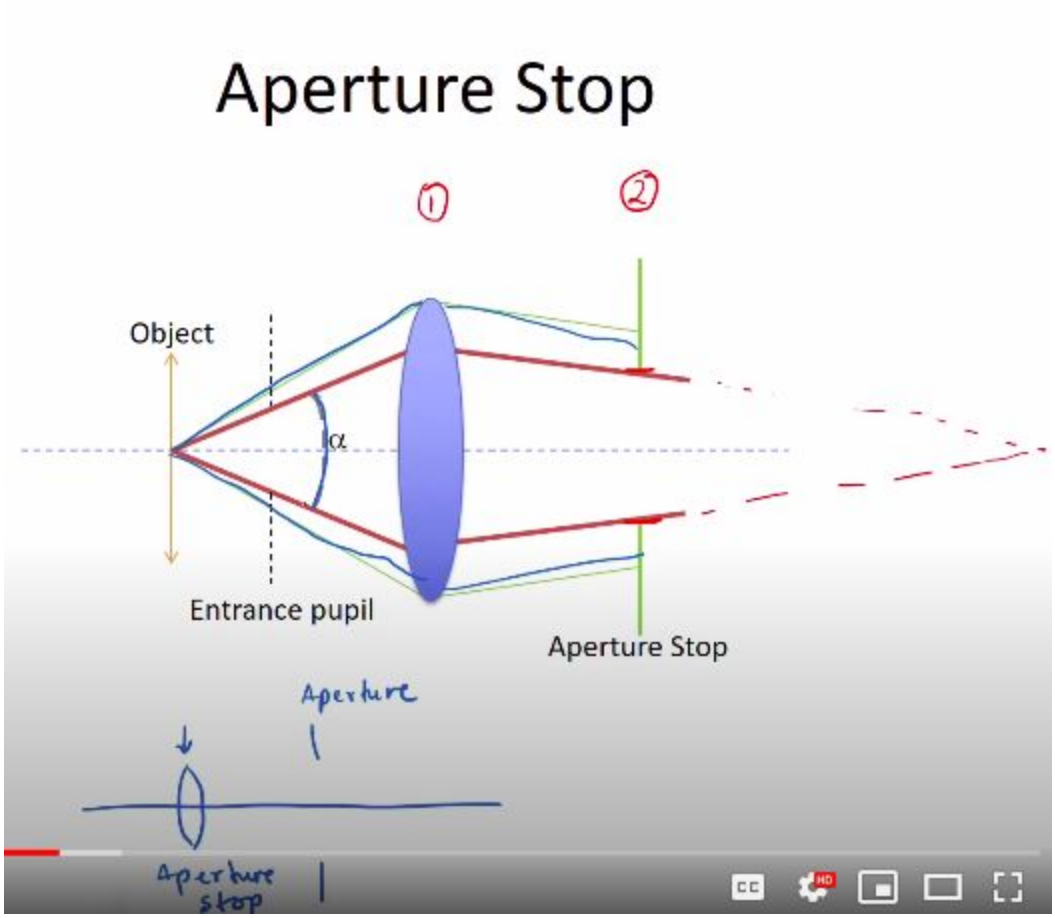
- Ray through centre of lens, and through front and back focal points can always be traced but may not actually travel through system
- Types of apertures
  - Aperture Stop
  - Field Stop
  - Glare stop

Aperture Stop

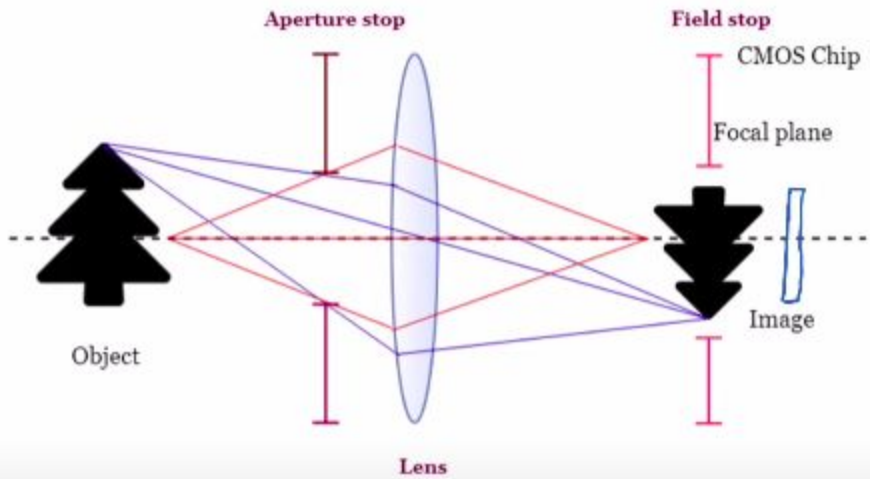


- The element that limits the cone of light in an optical system
- In OSLO, a surface is designated as the aperture stop with the initial AS
- A means that this surface is the aperture stop
- S means that the size of the surface is governed by a "paraxial solve," : size will be adjusted to accommodate (approximately) both axial and off-axis beams without truncation

entrance beam radius


Aperture Stop
1
2
Object
α
Entrance pupil
Aperture Stop
Aperture
Aperture stop

Simple Definitions

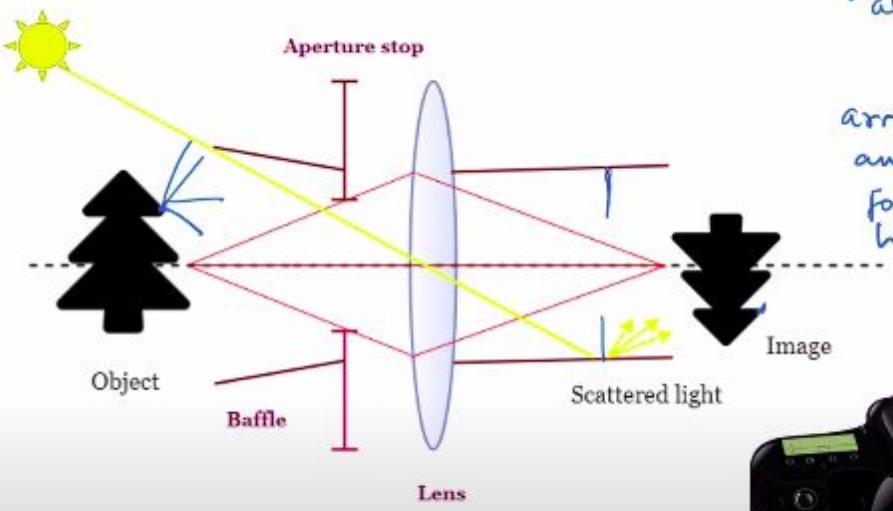


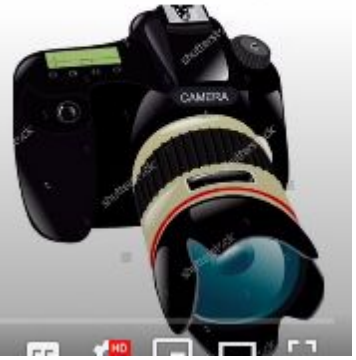
CCD
CMOS
sensor at image plane

- Field stop determines whether edges of image are actually captured
- Aperture stop limits cone of light from an object point
  - Brightness
  - Depth of focus/field
  - System resolution
  - Certain aberrations

Glare stops and baffles

For imaging
→ good focusing ability
↓
all the rays arriving at any pt in focus must have travelled the same optical path length

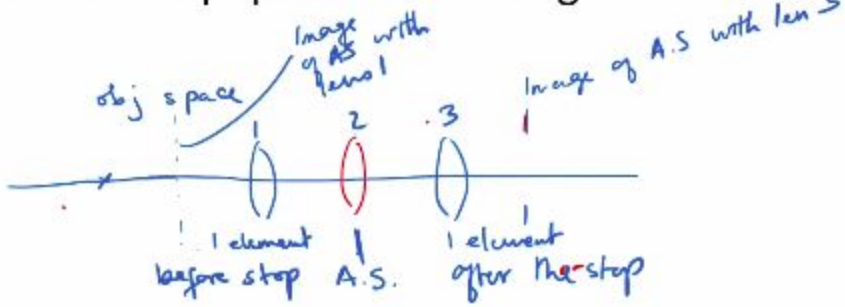




- Bright lights will interfere with image

Some definitions

- Pupil
  - Used to determine whether a ray traverses the entire system or not
- Entrance pupil
  - Image of aperture stop as seen from an axial point in object space through elements preceding stop
- Exit pupil
  - Image of A.S. see from axial point in image space
- No ray outside cone of pupils reaches image



Some definitions

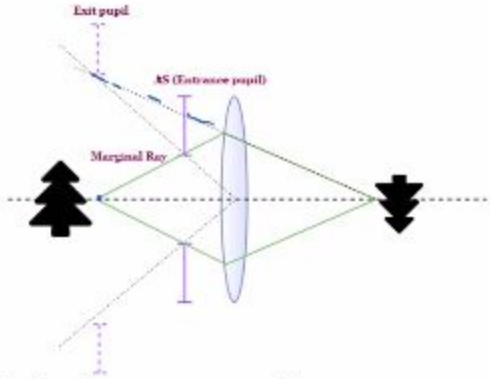
- Pupil
  - Used to determine whether a ray traverses the entire system or not
- Entrance pupil
  - Image of aperture stop as seen from an axial point in object space through elements preceding stop
- Exit pupil
  - entrance pupil, exit pupils
- No

Entrance and exit pupils of two camera lenses
Source: Carl Zeiss AG

Axial/Marginal Ray

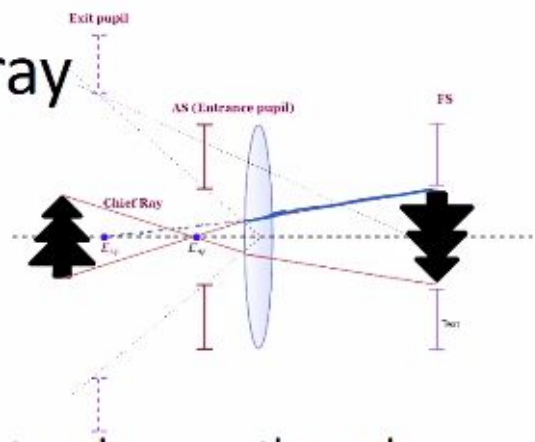
axial obj pt

outer edge of the entrance pupil



- Starts at the point where the object crosses the optical axis and touches the edge of the aperture stop of the system.
- Crosses (or appears to cross) the optical axis again at the locations where an image will be formed.
- The distance of the marginal ray from the optical axis at the locations of the entrance pupil and exit pupil defines the sizes of each pupil.

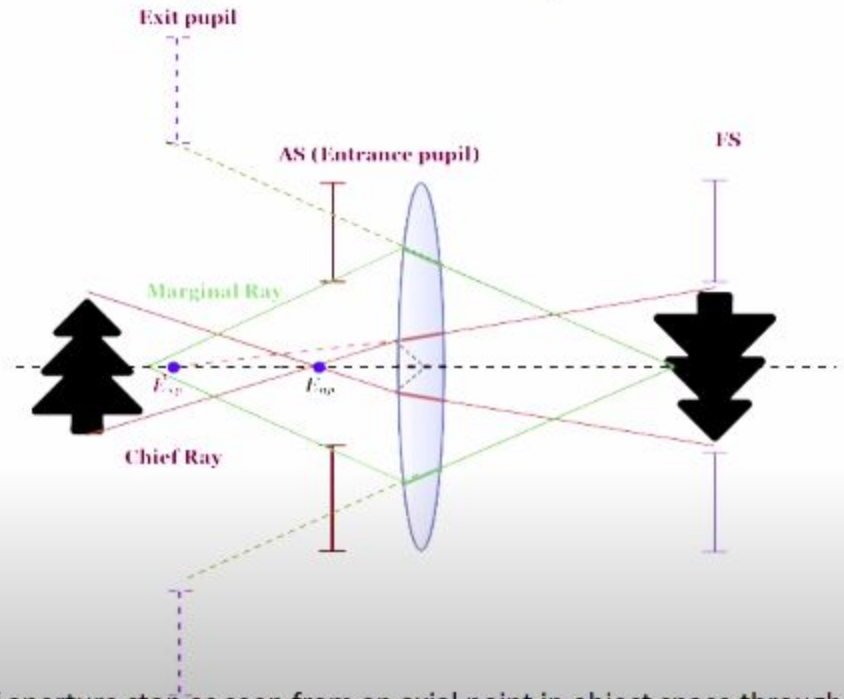
Principal ray or Chief ray



- Starts at the edge of the object and passes through the center of the aperture stop.
- Crosses (or appears to cross) the optical axis at the locations of the pupils.
- The distance between the chief ray and the optical axis at an image location defines the size of the image

3:55

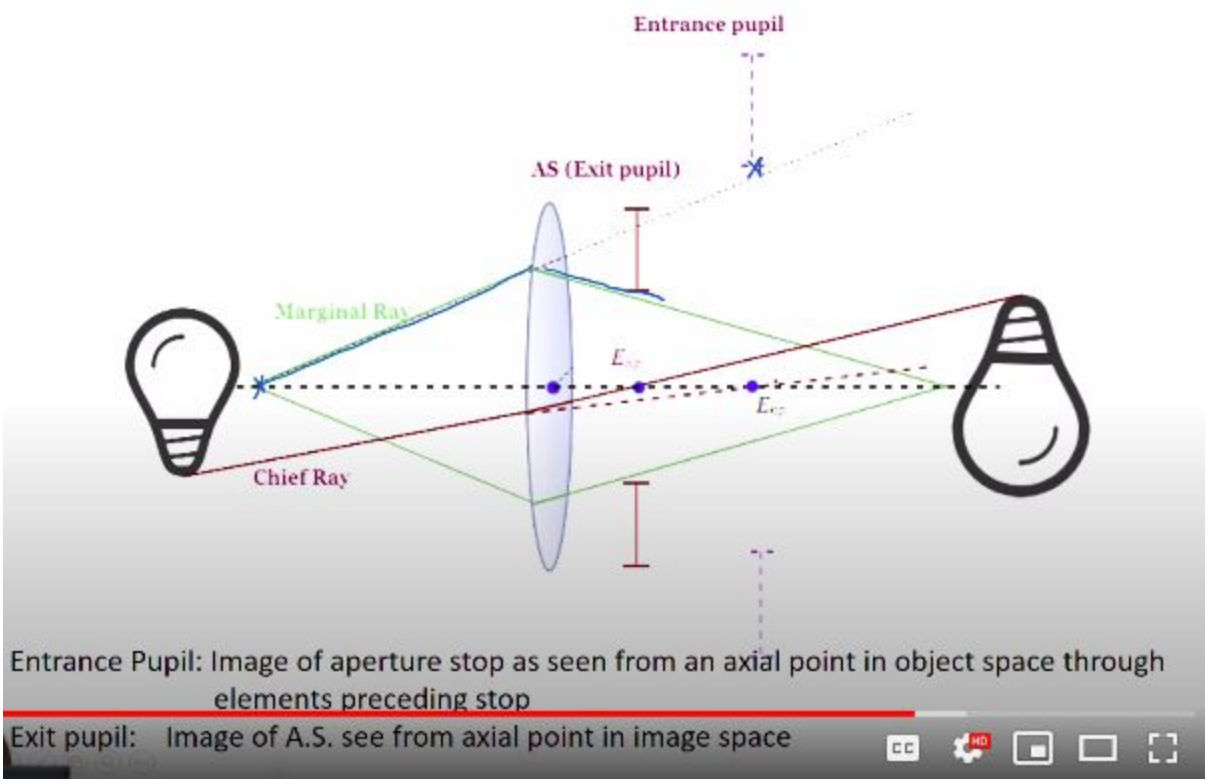
Front Aperture Stop



Entrance Pupil: Image of aperture stop as seen from an axial point in object space through elements preceding stop

Exit pupil: Image of A.S. see from axial point in image space

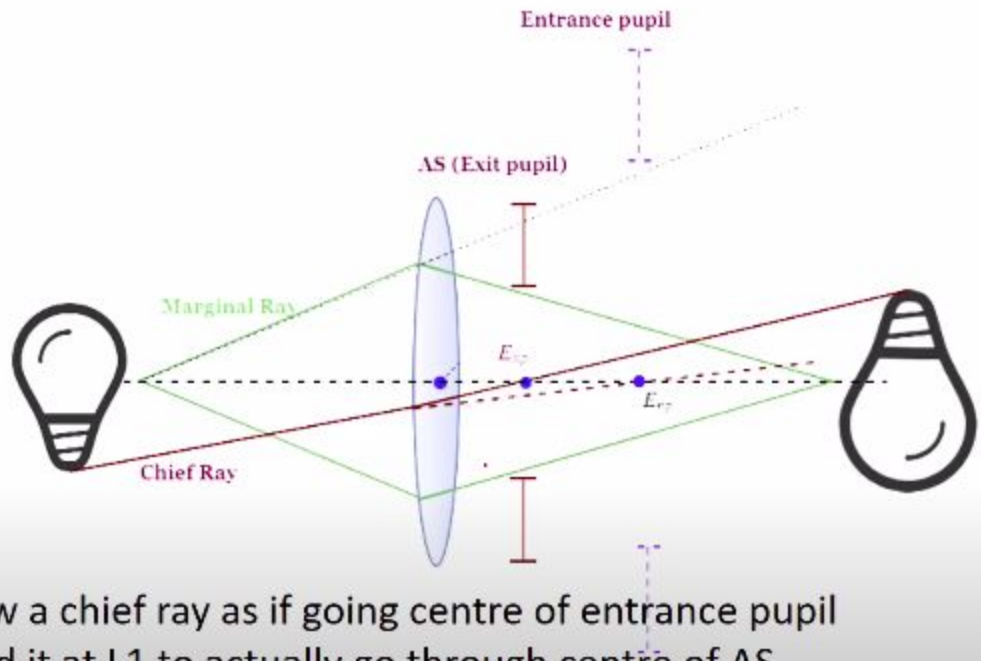
Back Aperture Stop



Entrance Pupil: Image of aperture stop as seen from an axial point in object space through elements preceding stop

Exit pupil: Image of A.S. see from axial point in image space

Tracing Rays



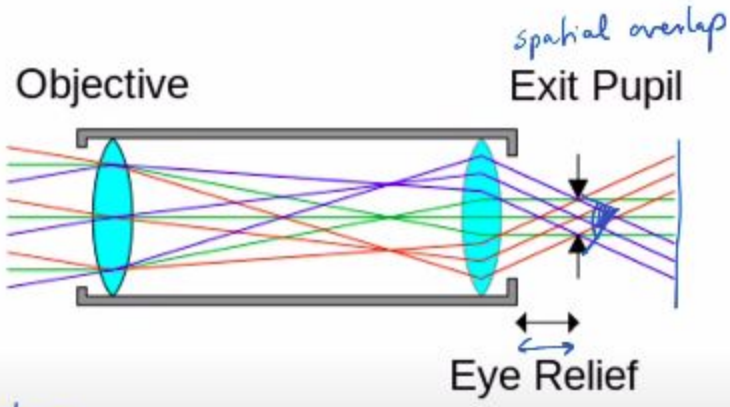
Draw a chief ray as if going centre of entrance pupil

Bend it at L1 to actually go through centre of AS

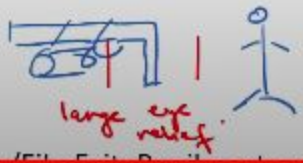
Draw marginal ray as if going through outer edge of entrance pupil

Bend it at L1 to actually go through outer edge of exit pupil

Exit Pupil – infinite conjugates



Optical system
: sight
attached to a rifle



://en.wikipedia.org/wiki/Exit_pupil#mediaviewer/File:Exit_Pupil_raytrace.s